LESSON 12.8
EVALUATING EXPRESSIONS USING THE ORDER OF OPERATIONS

LESSON INTRODUCTION

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers, including exponents and radicals.

LEARNING TARGETS

- I can solve problems involving the order of operations with rational numbers, including exponents and radicals.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents and absolute value.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How does the placement of grouping symbols impact the value of an expression?
- How do I evaluate multi-step problems involving the order of operations with grouping symbols?

LESSON NARRATIVE

Students continue to practice evaluating multi-step problems that have rational numbers, radicals, and exponents. In this lesson while evaluating expressions, students examine the impact of grouping symbols. Through engaging in the lesson activities, students will better understand that when the placement of grouping symbols changes the order which the operations are performed, the value of the expression will change.

COMPONENT SUMMARY

12.8.1 WARM-UP *(~5 MIN.)*

Find an approximate value for a real-world context

12.8.2 GUIDED PRACTICE *(10-15 MIN.)*

Evaluate expressions

12.8.3 COLLABORATION *(20-30 MIN.)*

Analyze student work and evaluate expressions having grouping symbols

12.8.4 TRY IT *(10-15 MIN.)*

Determine how the placement of parentheses can impact the value of an expression

12.8.5 WRAP-UP *(~5 MIN.)*

Find the value of an expression involving rational numbers, radicals, and exponents

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- Consider strategic grouping throughout the lesson. Determine if pulling small groups during the lesson are needed to ensure students have the opportunity to be successful on 12.8.5 Wrap-Up.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not use the reciprocal of the given fractions when evaluating negative exponents.
- Students may evaluate expressions according to the operations that come first, left to right, instead of applying the order of operations.

THINGS TO CONSIDER

- 12.8.1 Warm-Up involves converting measurements within the metric system. Consider having metric conversion charts available to assist students with equivalent metric measurement.
- This benchmark will be assessed as no calculator on the Florida Statewide Assessment (FSA).
- Students should not use a calculator to complete 12.8.5 Wrap-Up.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

In 12.8.3 Collaboration, question 1 asks students to evaluate an expression for different values of the variable. Consider using this routine for question 1. Student responses can be posted around the room. Once displayed, students can search the answers for negative values, if any, and can discuss how the negative values were found.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Mathematical fluency covers being flexible and precise while carrying out calculations. By providing students with multiple opportunities to execute procedures and perform calculations, 12.8.5 Wrap-Up allows students to maintain flexibility and accurately perform the order of operations with rational numbers, exponents, and radicals.

DIFFERENTIATE INSTRUCTION

Consider allowing students to check their final answers using a calculator but not to perform calculations through the lesson. This will give students experience with inputting expressions involving rational numbers, exponents, radicals, and grouping symbols into a scientific calculator.

Order of operations can be a struggle for students throughout high school mathematics courses. Having conceptual understanding and mathematical fluency of the concept is vital for future math courses. Monitor student progress during the lesson and determine if any students need small, group learning support. If it is determined there are students in need of additional support (use observational data from the lesson as well as formative assessment data from 12.8.5 Wrap-Up), create a plan for support, which should include strategic grouping of students, during Lessons 9 and 10.

12.8.1 WARM-UP: BETWEEN TWO NUMBERS

TEACHER MOVES

Based on the needs of your students determine if pairing students for 12.8.1 Warm-Up or allowing them to engage in a Think-Pair-Share would be a productive strategy to support learners who may not know where to begin with answering the question.

12.8.1 Warm-Up: Between Two Numbers

An octopus is capable of laying $56{,}000$ eggs at one time. Each egg is about the size of a single grain of rice, approximately 0.029 grams (g).

An ostrich, on the other hand, can lay 10 eggs at a time with each egg weighing as much as 1.5 kilograms (kg).

$$1 \text{ g} = 0.001 \text{ kg}$$

1. Approximately how many octopus' eggs would weigh the same as one ostrich's egg?
   $51{,}724$ *octopus' eggs*

12.8.2 Guided Practice: Evaluating Expressions

Find the value of each expression. Show your work.

1. $\left(\frac{1}{2} \cdot 4\right)^{-3} - \frac{3}{4} \cdot 12$

$= (2)^{-3} - \frac{3}{4} \cdot 12$

$= \frac{1}{2^3} - \frac{3}{4} \cdot 12$

$= \frac{1}{8} - \frac{3}{4} \cdot 12$

$= \frac{1}{8} - 9$

$= \frac{1}{8} - \frac{72}{8}$

$= -\frac{71}{8} = -8\frac{7}{8}$

2. $\sqrt[3]{-125} - 3 \cdot 5^2 + \left(\frac{1}{8}\right)^{-1}$

$= -5 - 3 \cdot 5^2 + \left(\frac{1}{8}\right)^{-1}$

$= -5 - 3 \cdot 25 + \left(\frac{1}{8}\right)^{-1}$

$= -5 - 3 \cdot 25 + 8$

$= -5 - 75 + 8$

$= -80 + 8$

$= -72$

3. $z \cdot \sqrt{21z + (6^2 - 1)}$, where $z = \frac{2}{3}$

$= \frac{2}{3} \cdot \sqrt{21 \cdot \frac{2}{3} + (6^2 - 1)}$

$= \frac{2}{3} \cdot \sqrt{21 \cdot \frac{2}{3} + (36 - 1)}$

$= \frac{2}{3} \cdot \sqrt{21 \cdot \frac{2}{3} + 35}$

$= \frac{2}{3} \cdot \sqrt{14 + 35}$

$= \frac{2}{3} \cdot \sqrt{49}$

$= \frac{2}{3} \cdot 7$

$= \frac{14}{3}$

$= 4\frac{2}{3}$

12.8.2 GUIDED PRACTICE: EVALUATING EXPRESSIONS

TEACHER MOVES

Listing the steps students take to solve each problem will provide an opportunity for students to practice the use of a wide variety of mathematical language. Have the students write out the steps they perform as they evaluate each expression.

ENRICHMENT

Before completing question 2, ask students questions to assess their understanding of the process to evaluate negative exponents, such as:

- Will a fraction raised to the power of -1 result in the same answer as a fraction raised to the power of 1?
- What is a reciprocal?
- If the decimal equivalent of the fraction were used instead, would the final answer be the same?

12.8.3 COLLABORATION: ANALYZING STUDENT WORK

DIFFERENTIATE INSTRUCTION

Collaboration and discussion are imperative throughout this section. Consider providing sentence starters to guide student discussion to target key learning opportunities as students work collaboratively through this section.

- Question 1a: I think Veronica/Roscoe is correct because…
- Question 1d: Veronica/Roscoe is correct because the product of squared numbers is always…

TEACHER MOVES

Notice that questions 1c and 1e do not require a written response, students are expected to discuss their thoughts verbally with a partner. Consider how to pair students and listen for responses that indicate understanding:

- Question 1c – I notice the value of the expression is positive no matter what the value of n is. I notice that the value of the expression is the same when n is 100 or 144.
- Question 1e – Removing the parentheses will change which values are possible for the expression because the exponent now only applies to 11, not the entire expression. This means the expression could have negative values as well.

12.8.3 Collaboration: Analyzing Student Work

1. Veronica and Roscoe were discussing how the value of the expression $(\sqrt{n}-11)^2$ would change for different values of n. Veronica said it is possible for the value of the expression to be negative, but Roscoe says that's impossible.

   a. Before performing any calculations, explain who you think is correct.
   Answers will vary.

   b. Work with your partner to evaluate the expression for each value of n given in the table. Show your work in the space provided.

Value of n	Show Your Work	Value of $(\sqrt{n}-11)^2$
25	$(\sqrt{25}-11)^2$ $(5-11)^2$ $(-6)^2$ 36	36
100	$(\sqrt{100}-11)^2$ $(10-11)^2$ $(-1)^2$ 1	1
144	$(\sqrt{144}-11)^2$ $(12-11)^2$ $(1)^2$ 1	1

   c. With your partner, discuss what you notice about the value of the expression for each value of n.

   d. Explain whether Veronica or Roscoe is correct.
   Answers will vary. Roscoe is correct because the products of squared numbers are always positive.

   e. Discuss with your partner if removing the parentheses, so the expression is $\sqrt{n}-11^2$, would change which values are possible for the expression.

Work with your partner to complete the following.

2. Liesel evaluated the expression $\frac{(-11+2^3)^2}{\sqrt{4^3}} - \frac{1}{4}$. Her work is shown, beginning with the original expression.

$\frac{(-11+2^3)^2}{\sqrt{4^3}} - \frac{1}{4}$	
$\frac{(-11+8)^2}{\sqrt{4^3}} - \frac{1}{4}$	
$\frac{(-3)^2}{\sqrt{4^3}} - \frac{1}{4}$	
$\frac{-9}{\sqrt{4^3}} - \frac{1}{4}$	$\frac{9}{\sqrt{4^3}} - \frac{1}{4}$
$\frac{-9}{4} - \frac{1}{4}$	$\frac{9}{8} - \frac{1}{4}$
$-\frac{10}{4}$	$\frac{9}{8} - \frac{2}{8}$
-2.5	$\frac{7}{8}$

Liesel made errors when evaluating the expression.

a. Circle where Liesel made errors.

b. Correct her work in the space provided beside Liesel's work.

c. Write Liesel a short note with advice on how to avoid similar mistakes in the future.
Answers will vary. Liesel, remember that negative numbers squared result in positive products. Also, remember that any number raised to the third power means to multiply the number itself three times.

3. Deshawn is evaluating the expression $-4(\sqrt{36}+21)$. Instead of first performing operations within the parentheses, he multiplied the -4 by both terms in the parentheses, as shown.

$$-4(\sqrt{36}+21) = (-4 \cdot \sqrt{36}) + (-4 \cdot 21)$$

a. Which property of operations did Deshawn use?
The Distributive Property of Multiplication over Addition

b. Evaluate both expressions Deshawn set equal to each other using the order of operations.

$-4(\sqrt{36}+21)$	$(-4 \cdot \sqrt{36}) + (-4 \cdot 21)$
$-4(6+21)$	$(-4 \cdot 6) + (-4 \cdot 21)$
$-4(27)$	$(-24) + (-84)$
-108	-108

c. Explain how the values of the two expressions compare.
Answers will vary. The value of the two expressions are the same.

12.8.3 COLLABORATION: ANALYZING STUDENT WORK (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider allowing students alternatives to writing a note to Liesel. An acceptable alternative would be for partners to explain verbally to a different classmate the advice they would give Liesel. This will give an opportunity for students to engage in conversation with another classmate and be able to discuss if they would have given different advice.

TEACHER MOVES

During the debrief for question 3c, highlight some of the nuances students may observe as they evaluate multi-step numerical expressions. For example, while performing operations within grouping symbols is typically done before multiplying, in a situation where a value is being multiplied by a group of values, the result is the same whether the multiplication is done first or last, as long as the multiplication is distributed to all parts of the expression in the group. This directly relates to what students saw when finding equivalent algebraic expressions. When the expression can't be evaluated (i.e., it's algebraic), multiplying each part of the expression by the common factor results in an equivalent expression...so a numerical expression can be evaluated in the same way.

12.8.4 TRY IT: THE IMPACT OF GROUPING

QUESTIONING

Pose another expression with the parentheses around the entire expression except for the negative sign. Encourage students to use a graphic organizer or focused notetaking to evaluate the expressions in question 1a. By using focused notetaking, students will have organized visuals to be able to reference.

DIFFERENTIATE INSTRUCTION

Question 2 could also be done as a four corners activity where each group evaluates one of the expressions and then students discuss questions 2b and 2c with their partner or group after seeing the results of each.

TEACHER MOVES

This is a discussion opportunity. Students do not need to write an explanation here. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

12.8.4 Try It: The Impact of Grouping

1. Chandler says that when following the order of operations, the two expressions below are equivalent and the parentheses are not needed.

$$(-3\cdot\sqrt{121})+19=-3\cdot\sqrt{121}+19$$

   a. Explain whether Chandler is correct. Justify mathematically.
   Answers will vary. Chandler is correct because the value of both expressions is -14.

   b. Explain whether the parentheses would change the value of the expression if they were around $\sqrt{121}+19$ instead.
   Answers will vary. The parentheses would change the value of the expression to -90 because the order of operations states that parentheses are evaluated before multiplication.

2. Four expressions are shown. One does not include any grouping symbols, while the others have parentheses placed around different parts of the expression.

Expression A	Expression B
$\left(\frac{1}{5}\cdot 5-\frac{3}{4}\right)\cdot\sqrt{64}^3$	$\frac{1}{5}\cdot\left(5-\frac{3}{4}\cdot\sqrt{64}\right)^3$
Expression C	**Expression D**
$\left(\frac{1}{5}\cdot 5-\frac{3}{4}\cdot\sqrt{64}\right)^3$	$\frac{1}{5}\cdot 5-\frac{3}{4}\cdot\sqrt{64}^3$

   a. Work with your partner to evaluate each expression. Show your work in the space provided.

Expression	Value
A	128
B	$-\frac{1}{5}$
C	-125
D	-383

$\left(\frac{1}{5}\cdot 5-\frac{3}{4}\right)\cdot\sqrt{64}^3$
$\left(\frac{1}{4}\right)\cdot\sqrt{64}^3$
$\left(\frac{1}{4}\right)\cdot 512$
128

$\frac{1}{5}\cdot\left(5-\frac{3}{4}\cdot\sqrt{64}\right)^3$
$\frac{1}{5}\cdot(-1)^3$
$\frac{1}{5}\cdot -1$
$-\frac{1}{5}$

$\left(\frac{1}{5}\cdot 5-\frac{3}{4}\cdot\sqrt{64}\right)^3$
$\left(\frac{1}{5}\cdot 5-\frac{3}{4}\cdot 8\right)^3$
$(1-6)^3$
$(-5)^3$
-125

$\frac{1}{5}\cdot 5-\frac{3}{4}\cdot\sqrt{64}^3$
$\frac{1}{5}\cdot 5-\frac{3}{4}\cdot 8^3$
$\frac{1}{5}\cdot 5-\frac{3}{4}\cdot 512$
$1-384$
-383

   b. Discuss with your partner how the placement of parentheses can impact the value of an expression.

When the placement of grouping symbols changes the order in which the operations are performed, the value of the expression changes.

12.8.5 Wrap-Up: Ticket Out the Door

1. Without a calculator, find the value of the expression. Show your work.

$\sqrt{\frac{10^2 - 1}{11}} - 0.75 + 2^{-2}$

$= \sqrt{\frac{100 - 1}{11}} - 0.75 + 2^{-2}$

$= \sqrt{\frac{99}{11}} - 0.75 + 2^{-2}$

$= \sqrt{9} - 0.75 + 2^{-2}$

$= 3 - 0.75 + 2^{-2}$

$= 3 - 0.75 + \frac{1}{2^2}$

$= 3 - 0.75 + \frac{1}{4}$

$= 2.25 + \frac{1}{4}$

$= 2.25 + 0.25$

$= 2.5$

12.8.5 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

Proficiency on 12.8.5 Wrap-Up includes:

- Fluency in performing the order of operations with rational numbers, exponents, and radicals
- Fluency in rewriting numerical bases with negative exponents using positive exponents
- Fluency in precise and accurate calculations